

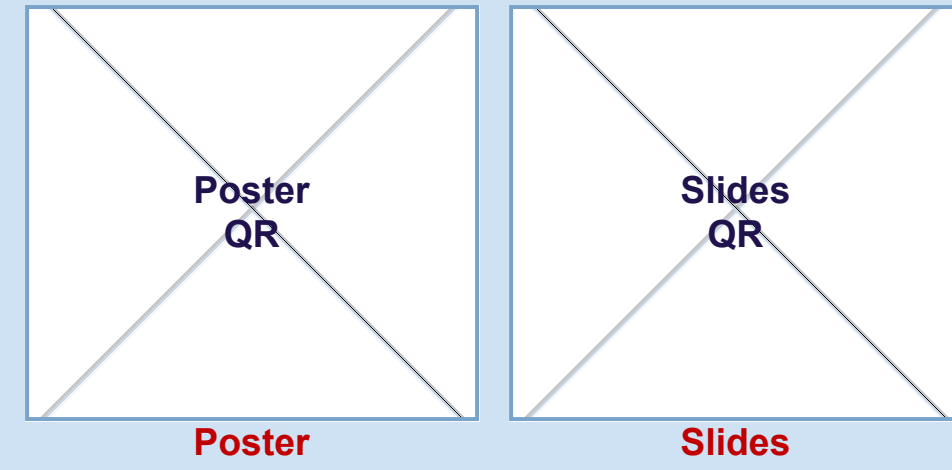


DeXOR: Enabling XOR in Decimal Space for Streaming Lossless Compression of Floating-point Data

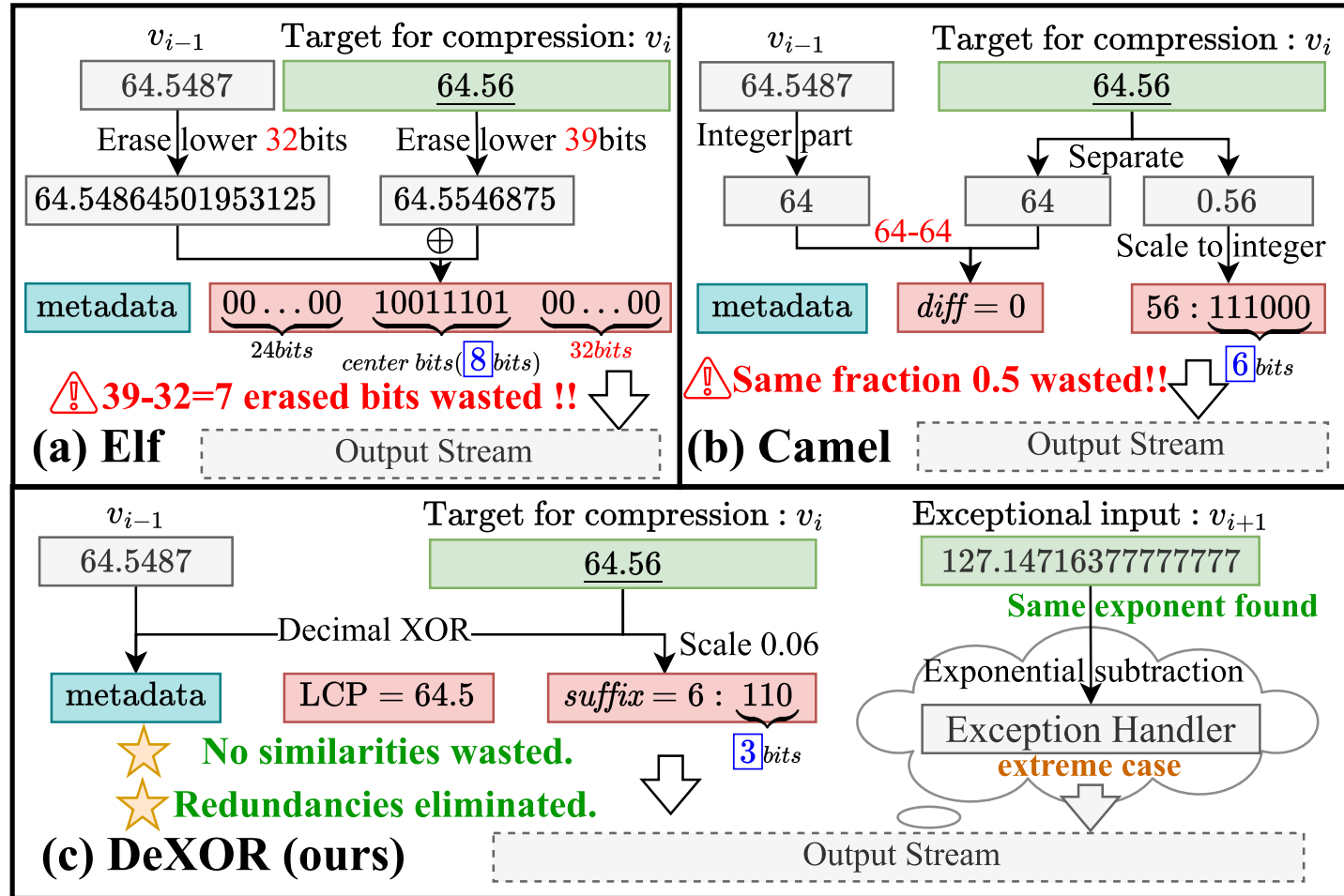
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Email: chuanyi.lv@zju.edu.cn Repository: github.com/SuDIS-ZJU/DeXOR



1. Introduction



Challenge 1: Reconstruction Error Control.

Preserve exact recovery.

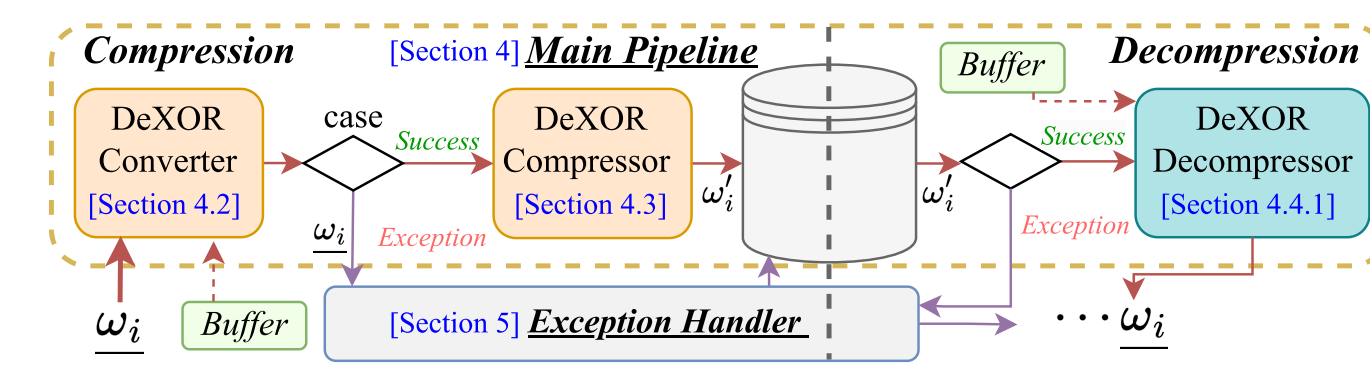
Challenge 2: Computational Overhead.

Reduce prefix, suffix, and metadata costs.

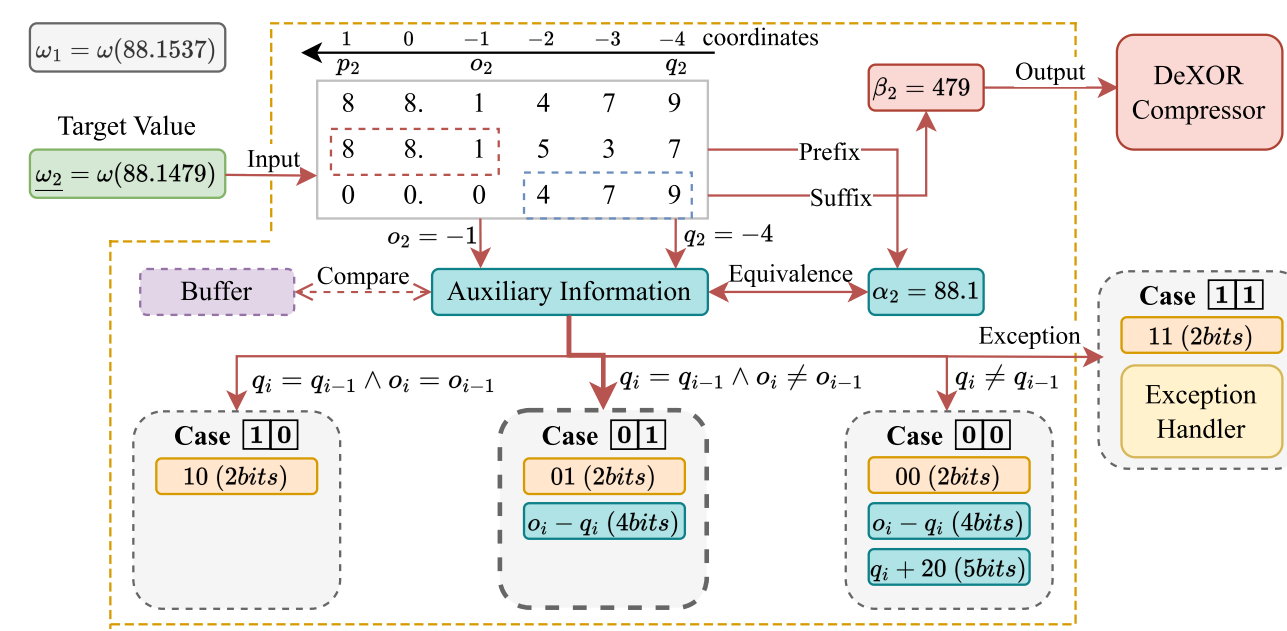
Challenge 3: Robustness in Extreme Cases.

Handle high-precision and non-smooth streams.

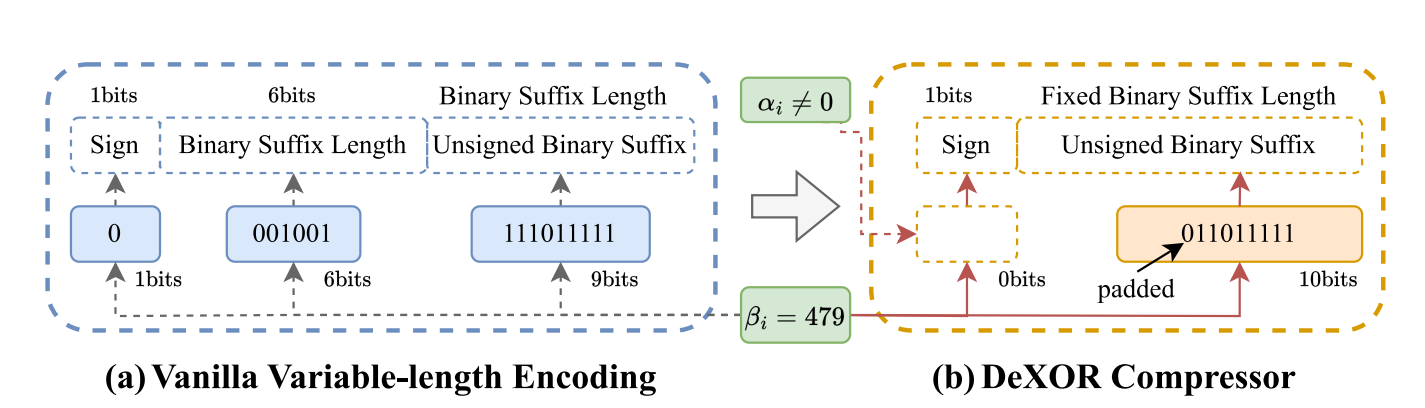
2. DeXOR Framework and Techniques



Overall framework. Converter and Compressor handle regular inputs, with difficult values passed to the Exception Handler.

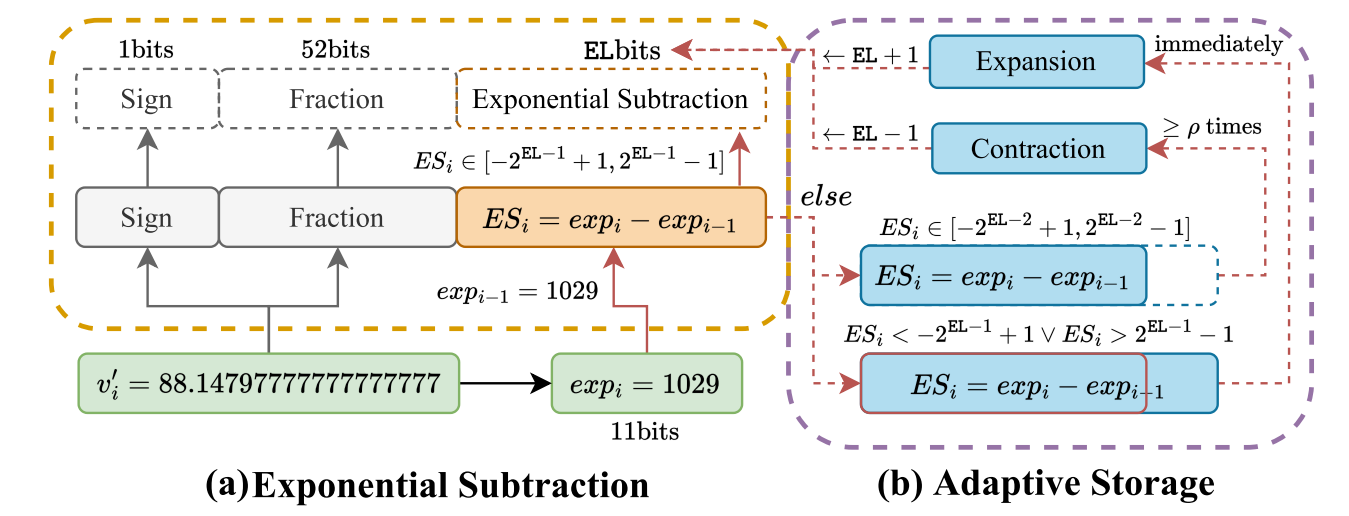


Decimal XOR. Decimal XOR removes the shared prefix and converts the suffix into an integer.



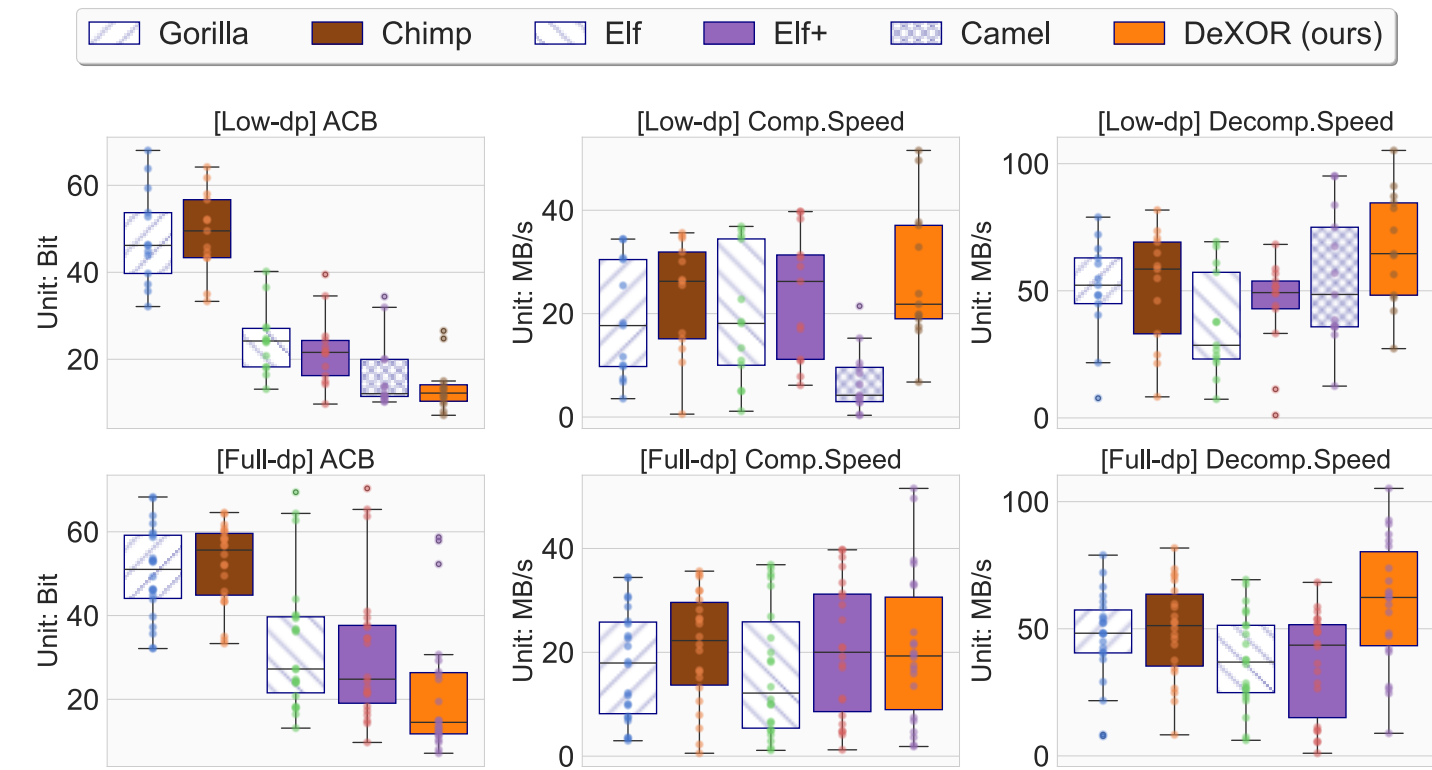
Suffix encoding. Sign consistency removes sign and length fields.

Error tolerance. After scaling, $\Delta = 10^{-6}$ compensates rounding errors during metadata extraction.



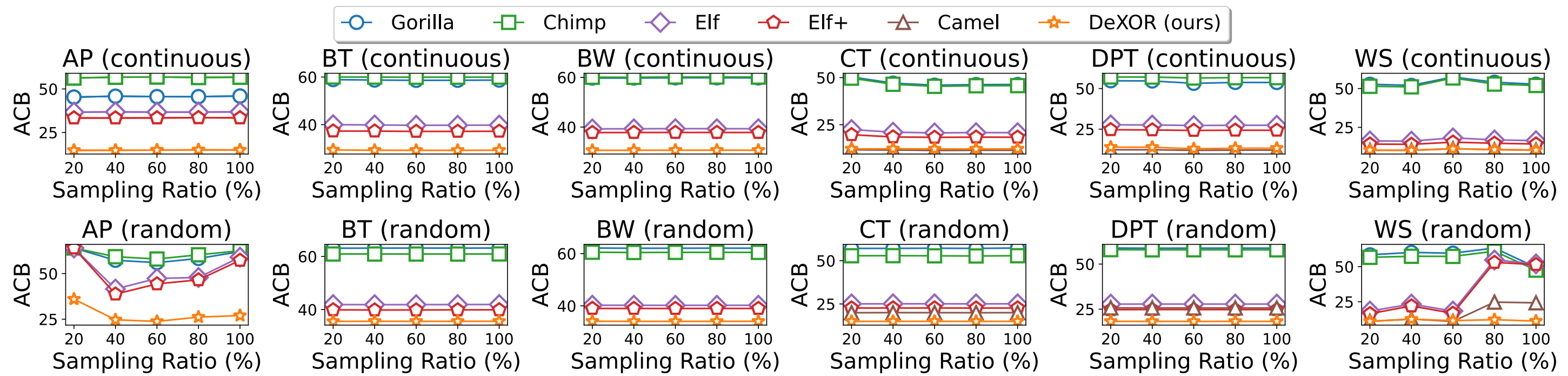
Exception Handler. The binary path prevents expansion and ensures bitwise recovery.

3. Experimental Studies

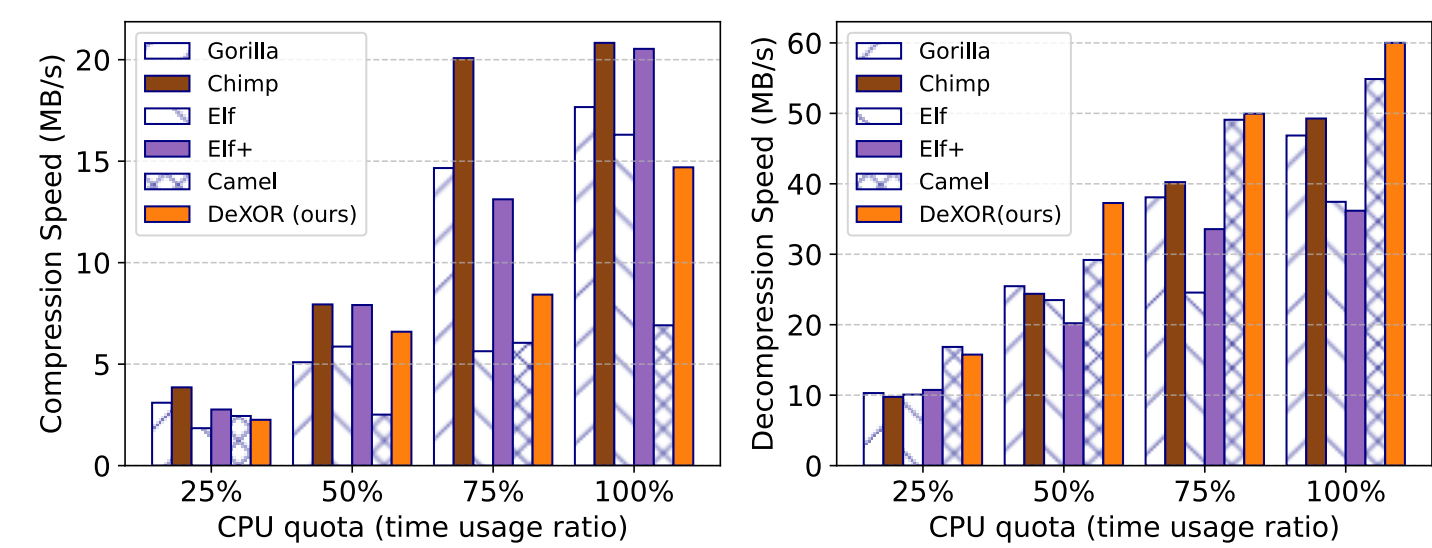


Overall.

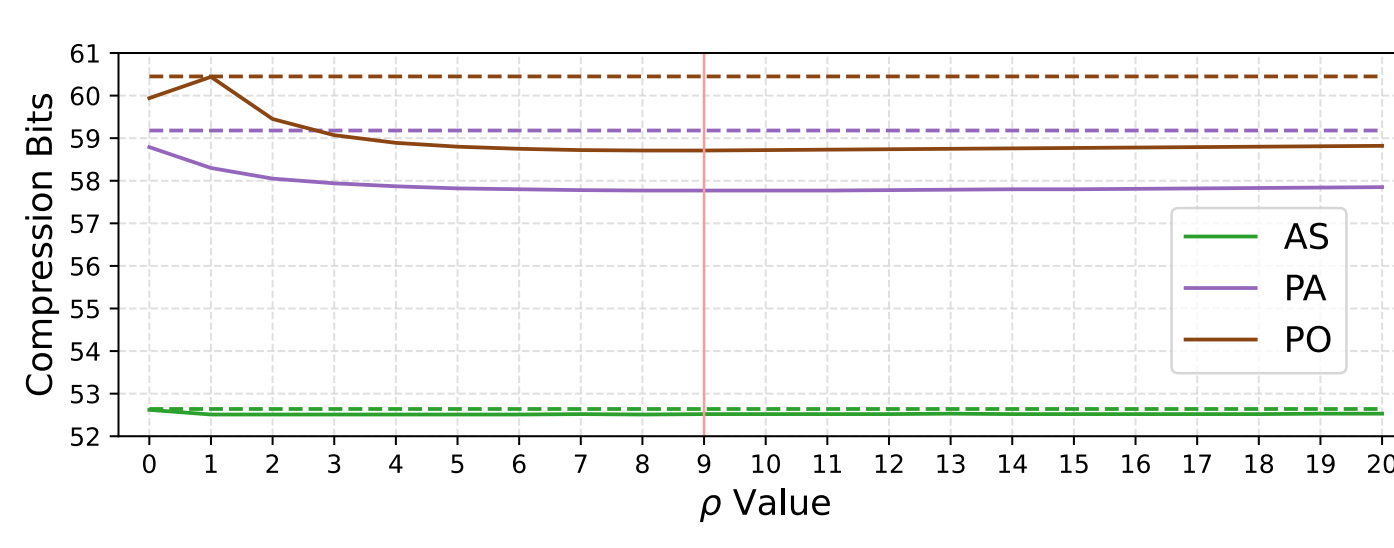
Across 22 datasets, DeXOR improves compression ratio by **15%** and decompression speed by **20%**.



Sampling variability. DeXOR consistently achieves the lowest ACB with little variation across continuous and random sampling ratios.



CPU scaling. At 50%–100% CPU quota, DeXOR leads in compression and decompression speed.



Parameter sensitivity. Default $p = 8$ stays near optimal on high-precision datasets without tuning.

System integration.

In IoTDB, DeXOR improves compression by up to **4.5x** while retaining ingestion throughput and reducing query latency.

Ablation.

Without Decimal XOR, average ACB rises by **19.74%**. The Exception Handler prevents expansion on high-precision inputs.

Vector data.

Dimension-wise DeXOR outperforms Gorilla on SIFT and Wine Quality across all reported metrics.

4. Beyond Lossless: E-DeXOR

Updated problem. SEC permits $|\hat{v}_i - v'_i| \leq 10^\epsilon$ when exact recovery is unnecessary.

Challenge 1: Reconstruction Error Control.

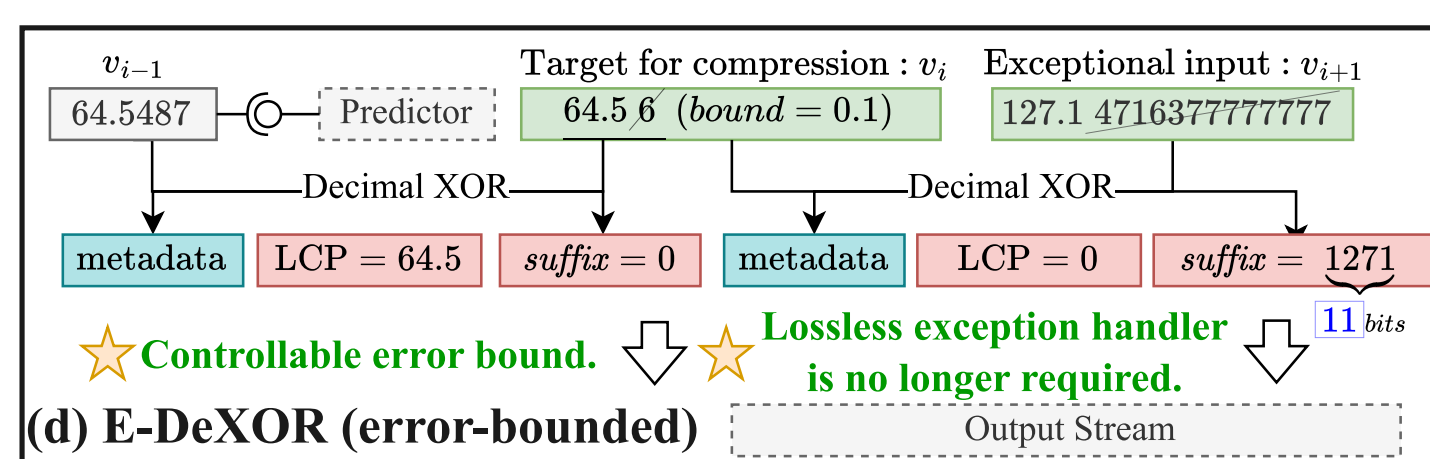
Keep reconstruction within the user-specified error bound.

Challenge 2: Computational Overhead.

Reduce data and metadata costs.

Challenge 3: Robustness in Extreme Cases.

Avoid costly exceptions for extreme tails and rounding artifacts.



Solution 1: Bound-aware truncation.

Truncate the low-order suffix according to the decimal precision bound.

Solution 2: Compact metadata.

Use 1-bit case encoding and precision-anchored metadata.

Solution 3: Exception absorption.

Absorb extreme tails and most rounding artifacts without a separate handler.

Expected effect. Higher and more robust compression with higher throughput under a controllable error bound.